

UNIT 5: Simple Machines

Subject: Physics

Grade: Grade 9

Core idea: Making work easier.

Types of Simple Machines

Lever, pulley, inclined plane, wheel and axle, and screw with practical classroom examples.

Mechanical Advantage and Velocity Ratio

Understand MA and VR to compare effort, load, and movement in machines.

Efficiency of Machines

Explain why no real machine is 100 percent efficient due to losses such as friction.

Real-Life Applications

Identify simple machines in tools, transport, construction, and household devices.

Mastery Goal

Analyze how machines reduce effort and improve efficiency.